SIMULATING THE SILENT THREAT: POPULATION DYNAMICS UNDER INVASIVE PRESSURE AND ENVIRONMENTAL SHOCKS LEADING TO LOCAL EXTINCTION

Heury Ferr
Department of General Biology
Institute of Biological Sciences
74 Esperança ave., Samambaia Campus, Goiânia, Brasil, 690900
heuryferr@gmail.com

Summary

This study investigates the synergistic effects of invasive species and environmental shocks on the local extinction risk of native species, a topic of increasing concern in the context of global change. Using a multi-species Lotka-Volterra competition model, it is possible to simulate the population dynamics of one invasive species and three native species with varying degrees of niche overlap to capture the heterogeneity of ecological interactions. Our results demonstrate that invasive species suppress native populations through direct competition, reducing their equilibrium abundances and, in several cases, driving them to local extinction. This effect is particularly pronounced for species with high niche overlap, which face the strongest competitive exclusion. In addition, it is possible to show that the reduced abundances of native species make them disproportionately vulnerable to both recurring environmental fluctuations and rare catastrophic perturbations. Such shocks can push already suppressed populations across extinction thresholds that they would likely withstand in the absence of invasive competitors. By combining these two stressors – chronic biological invasions and acute habitat disturbances – our model reveals a dual-threat mechanism that accelerates biodiversity loss. These findings underscore the importance of considering not only the direct ecological impacts of invasive species but also their role in amplifying the sensitivity of native communities to environmental variability. From a conservation perspective, this highlights the urgent need for strategies that integrate management of invasive populations with efforts to buffer ecosystems against shocks, thereby improving the resilience of native biodiversity in an era of intensifying human-driven change.

Keywords: invasive species, local extinction, environmental shocks, competitive suppression, biodiversity loss.

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1. Introduction

The escalating global loss of biodiversity is a defining challenge of the 21st century, with invasive species recognized as one of the primary drivers of species decline and extinctions worldwide [1]. By altering native ecosystems and outcompeting local flora and fauna, invasive species can fundamentally change community structure and function [2, 3].

The intensity of this competitive impact is largely determined by the degree of niche overlap between the invader and native species [4]. When two species share similar resource requirements, the principle of competitive exclusion suggests that one may outcompete and displace the other, potentially leading to local extinction [5].

While the direct competitive effects of invasive species are well-documented, their interaction with other drivers of biodiversity loss, such as environmental stochasticity, remains a critical area of research [6]. A population suppressed by chronic competitive stress may be more vulnerable to a subsequent, acute perturbation, such as a drought, disease outbreak, or wildfire. This phenomenon, which it is possible to term the "dual-threat mechanism", suggests that invasive species may not need to be competitively dominant to cause extinction [7]. Instead, they can weaken native populations to a point where they are unable to withstand a typical environmental shock, a fate they would likely avoid in a pristine, un-invaded state.

This study uses a system of ordinary differential equations (ODEs) to explore the dynamics of an ecosystem consisting of an invasive species and three native species with varying degrees of niche overlap. It is possible to simulate four distinct scenarios to investigate the population dynamics under different ecological pressures. First, it is possible to establish a baseline by simulating the native species in the absence of the invader. Second, it is possible to introduce the invader to observe its suppressive effect

on the native populations. Third, it is possible to simulate the impact of recurring environmental shocks on the invaded system. Finally, it is possible to model the effect of a single, catastrophic stochastic event to demonstrate the increased vulnerability of competitively suppressed native species. Through these simulations, let's aim to provide new insights into how the combined effects of chronic competition and environmental perturbations can accelerate the decline of native biodiversity.

2. Materials and methods

A computational model was developed in Python to simulate the population dynamics of one invasive species and three native species using a multi-species Lotka-Volterra competition framework. This model is a cornerstone of theoretical ecology, effectively describing the changes in population size over time due to carrying capacity limitations and interspecific competition. The rate of change for each species' population (x_i) is governed by the following system of differential equations. For the invasive species (x_1)

$$\frac{dx_1}{dt} = r_1 x_1 \left(1 - \frac{x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 + \alpha_{14}x_4}{K_1}\right). \quad (1)$$

For the native species [x_2, x_3, x_4]

$$\frac{dx_i}{dt} = r_i x_i \left(1 - \frac{x_i + \alpha_{i1}x_1 + \alpha_{i2}x_2 + \alpha_{i3}x_3 + \alpha_{i4}x_4}{K_i}\right), \quad (2)$$

where x_i – the population of species i, r_i – its intrinsic growth rate, and K_i – its carrying capacity. The term α_{ij} represents the competition coefficient, quantifying the per-capita competitive effect of species j on species i. To reflect the varying impacts of the invasive species, it is possible to assign different α_{i1} values for the native species: a high value for Native 1 (High Comp.), an intermediate value for Native 2 (Mod. Comp.), and a low value for Native 3 (Low Comp.), simulating distinct levels of niche overlap.

The model was implemented in Python using the scipy.integrate.odeint function for numerical integration over a time horizon of 200 units. The following scenarios were simulated:

Scenario 1. A baseline simulation with no invasive species ($r_1 = 0$), allowing the three native species to reach their stable equilibrium populations in a competition-only environment.

Scenario 2. The invasive species was introduced with an initial population of 10 individuals, allowing to observe its impact on the native populations and the resulting shifts in equilibrium.

Scenario 3. this scenario incorporated recurring environmental shocks. At specific time points, a sudden, proportional reduction was applied to all populations to simulate a disturbance event. For the stochastic shock moments [in times $t \in$ event_times]

$$x_i\left(t^+\right) = x_i\left(t^-\right) \times \left(1 - \sigma_i\right), \quad (3)$$

where t^- and t^+ represent the moments immediately before and after the shock; σ_i, is the shock's impact on species i.

Scenario 4. A stochastic event was modeled as an abrupt, single reduction in all populations at a specific time point, testing the resilience of the competitively suppressed native species.

3. Results

The simulations revealed distinct population dynamics across the four scenarios, highlighting the complex interplay between interspecific competition and environmental perturbations:

Scenario 1. Native Species dynamics ***(Fig. 1)****.* In the absence of the invasive species, the three native species (Native 1, Native 2, and Native 3) exhibited a classic competitive dynamic. After an initial period of fluctuation, all three species reached a stable, positive equilibrium.

Native Species dynamics without invasion. The population trajectories of three native species under baseline conditions (no invasive species) show their dynamics. Native Species 1 (high

competition) is outcompeted, dropping to fewer than 100 individuals, while Native Species 2 (moderate competition) stabilizes at around 680 individuals. Native Species 3 (low competition) maintains a stable population of approximately 800 individuals.

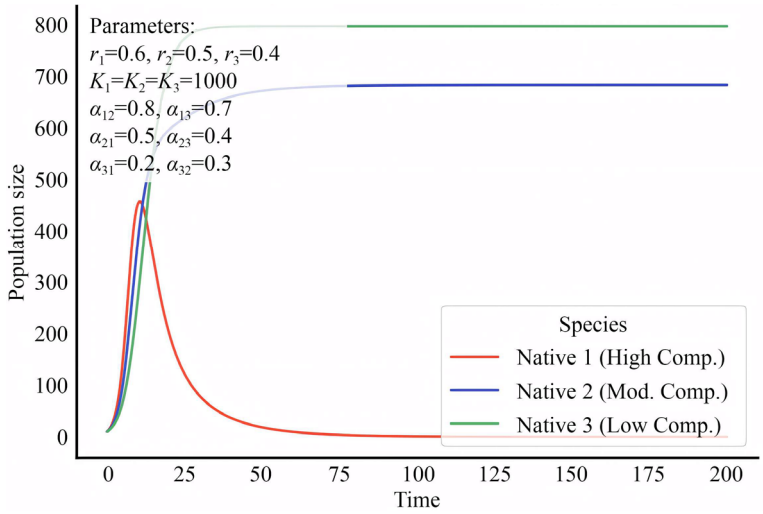


Fig. 1. Native species dynamics without invasion

Scenario 2. Impact of Invasive Species ***(Fig. 2).*** The introduction of the invasive species drastically altered the ecosystem's dynamics. The invasive species (solid red line) rapidly increases and stabilizes at approximately 1000 individuals, demonstrating its strong competitive ability. In contrast, Native 1 (High Comp.) is quickly outcompeted and driven to near extinction (dashed blue line). Native 2 (Mod. Comp.) experiences a significant decline but manages to persist at a much lower population size (dotted green line). Native 3 (Low Comp.) is the least affected, maintaining a population close to its carrying capacity, around 780 individuals (dash-dot green line).

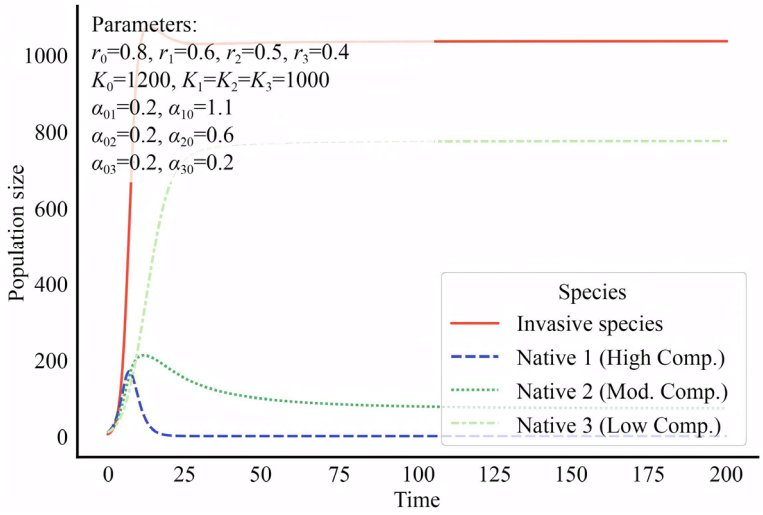


Fig. 2. Impact of invasive species

Scenario 3. Environmental shocks and species competition ***(Fig. 3).*** In this scenario, it is possible to investigate the effect of recurring environmental shocks on an ecosystem already impacted by the invasive species. The invasive species (solid red line) maintains a dominant population but experiences sharp drops after each shock, recovering quickly thereafter. Native Species 1 (high competition) remains at very low levels, unable to recover. Native Species 2 (moderate

competition) is significantly affected, with its population falling and stabilizing at even lower levels after each event. Native Species 3 (low competition) proves to be the most resilient, maintaining a high population and suffering only small, temporary reductions.

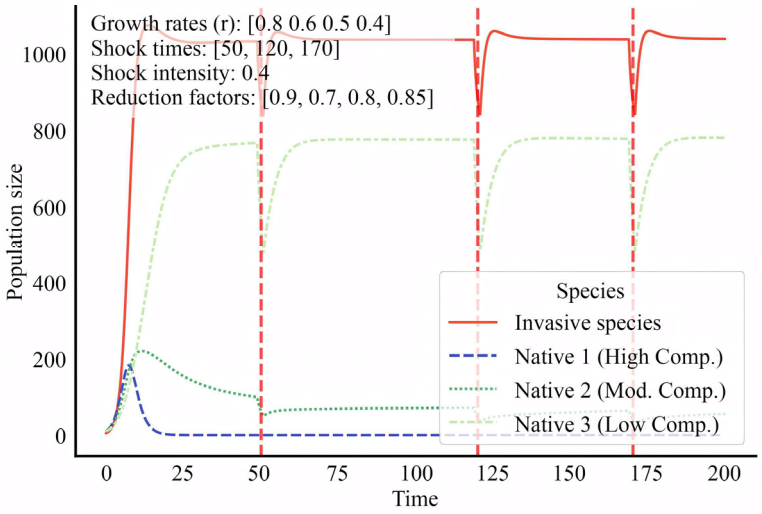


Fig. 3. Environmental shocks and species competition

Scenario 4. Stochastic Event Impacts ***(Fig. 4).*** This simulation, while visually similar to the previous scenario, underscores the vulnerability to single, abrupt stochastic events. The invasive species (solid red line) and Native 1 (high competition) maintain their respective dominant and suppressed population levels, showing no visible impact from the events. In contrast, Native 2 (moderate competition) and Native 3 (low competition) show minor fluctuations around their stable populations, suggesting that these species are more susceptible to subtle, random environmental noise. This scenario highlights a system that is resilient to stochastic events, unlike the vulnerability to large, periodic shocks seen in the previous scenario.

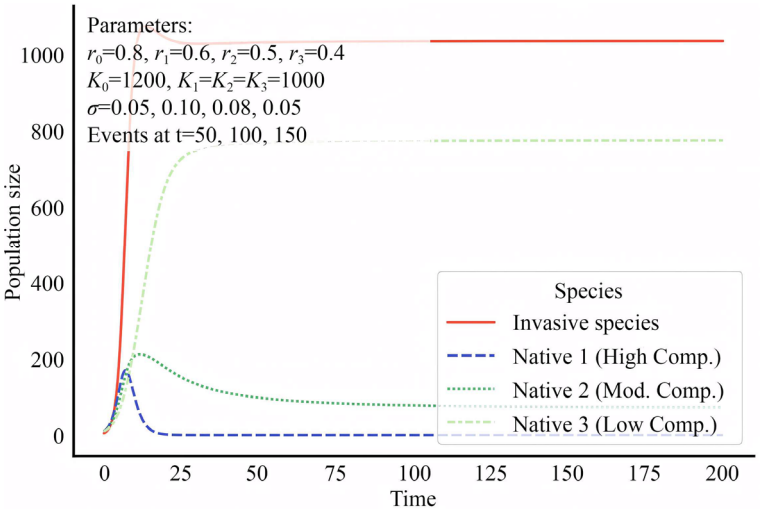


Fig. 4. Stochastic event impacts

4. Discussion

Our simulation results provide compelling evidence that invasive species not only suppress native populations through direct competition but also significantly increase their vulnerability to

environmental shocks. The model, based on the classic Lotka-Volterra equations, serves as a powerful framework for exploring the dual threats posed by biological invasions and habitat degradation.

The initial baseline simulation (Scenario 1) established that all three native species could coexist at stable, positive equilibrium populations. This scenario represents a healthy, balanced ecosystem. However, the introduction of the invasive species in Scenario 2 dramatically altered this dynamic. As predicted by the competitive exclusion principle, the native species with the highest niche overlap (Native 1) was completely outcompeted and driven to local extinction. This finding aligns with numerous empirical studies documenting the displacement of native species by invaders [8–11].

The core insight of this study emerges from Scenarios 3 and 4. In both simulations, the populations of the native species were chronically suppressed by the invasive species, maintaining them at levels far below their carrying capacity. This state of perpetual competitive pressure made them extremely sensitive to population reductions caused by external events. The recurring shocks in Scenario 3 and the singular stochastic event in Scenario 4 pushed these already weakened populations to the brink of extinction. This "double-whammy" effect-chronic competitive stress combined with acute environmental perturbations-provides a crucial mechanism for explaining accelerated rates of biodiversity loss in invaded ecosystems. These results are in agreement with many findings in the literature, which show through observational and experimental studies that invasive species often gain a competitive advantage under acute environmental perturbations, especially when they are more stress-tolerant than native species. For example, Heavy metal pollution and drought stress can accelerate the invasion of invasive alien species like *Amaranthus retroflexus*, as they show stronger competitiveness and stress intensity on their host plants [12].

In addition, invasive colonial ascidians show greater tolerance to environmental stress than native species, potentially favoring their success in coastal ecosystems impacted by climate change and bioinvasions [13]. Invasive tunicates, like Styela clava, perform better under extreme conditions like increased temperature and decreased salinity, potentially facilitating competitive shifts between native and invasive species in the future [14]. Moreover non-native marine invertebrates show greater tolerance to environmental stress and higher survival rates than native species, making them a useful criterion for screening profiles and risk assessment protocols [15].

Our findings suggest that conservation efforts must consider not only the direct impact of invasive species but also the subsequent increase in vulnerability of native populations. While a native species might be able to survive competition in a stable environment, it may not withstand the combined pressure of competition and climate change-induced extreme weather events, or disease outbreaks. The model's results highlight a critical need to understand how long-term ecological stress compromises the resilience of native communities.

5. Conclusion

This modeling study demonstrates that the introduction of an invasive species fundamentally shifts the stability and resilience of a native ecosystem. By suppressing the equilibrium populations of native competitors, an invader makes these species highly susceptible to local extinction from external environmental shocks, a threat that they could otherwise endure. Our results emphasize that an invasive species does not need to be a superior competitor to drive extinction; rather, its sustained presence can create a state of chronic vulnerability.

Future research should expand this model to include more complex dynamics, such as density-dependent mortality from shocks or the effects of varying resource availability. The findings underscore the urgency of preventative measures against biological invasions and the importance of incorporating these dual-threat dynamics into conservation planning to safeguard biodiversity in a rapidly changing world.

Conflict of interest

The authors declare that they have no conflict of interest in relation to this research, whether financial, personal, authorship or otherwise, that could affect the research and its results presented in this paper.

Financing

The study was performed without financial support.

Data availability

Data will be made available on reasonable request.

Use of artificial intelligence

The author confirms that he did not use artificial intelligence technologies when creating the current work.

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